MSG No. 136.502 $P4'_2/m'nm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,15,16]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,8,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[3,4,7,8]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[11,12,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,15,16]`
2	$[0,\ 0,\ -z]$	`[3,4,13,14]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8,9,10]`

Table 6: Wyckoff site: `4f`, site symmetry: `m'.2m'`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,3,14,16]`
2	$[-x,\ -x,\ 0]$	`[2,4,13,15]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[5,8,10,11]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[6,7,9,12]

Table 7: Wyckoff site: 4g, site symmetry: m'.m'2

No.	position	mapping
1	$[x,\ -x,\ 0]$	[1,4,14,15]
2	$[-x,\ x,\ 0]$	[2,3,13,16]
3	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[5,7,10,12]
4	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[6,8,9,11]

Table 8: Wyckoff site: 8h, site symmetry: 2..

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,2]
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	[3,4]
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[5,6]
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[7,8]
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,10]
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[11,12]
7	$\left[0,\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[\frac{1}{2},\ 0,\ z\right]$	[15,16]

Table 9: Wyckoff site: 8i, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-x,\ -y,\ 0]$	[2,13]
3	$[y,\ x,\ 0]$	[3,16]
4	$[-y,\ -x,\ 0]$	[4,15]
5	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[5,10]
6	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[6,9]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[7,12]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[8,11]

Table 10: Wyckoff site: 8j, site symmetry: ..m'

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$[-x,\ -x,\ z]$	[2,15]
3	$[x,\ x,\ -z]$	[3,14]
4	$[-x,\ -x,\ -z]$	[4,13]
5	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[5,11]

continued ...

Table 10

No.	position	mapping
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[6,12]
7	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[7,9]
8	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[8,10]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[y,\, x,\, -z]$	[3]
4	$[-y,\, -x,\, -z]$	[4]
5	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[5]
6	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[6]
7	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[8]
9	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[10]
11	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$[y,\, x,\, z]$	[16]